

```

#include <Servo.h>
#define ar analogRead
#define aw analogWrite

int x = A1, y = A0, z = A2;
int seuil = 10;
int xpos, ypos, zpos;
int xout, yout, zout;
float xmoy, ymoy, zmoy;

Servo motd;
Servo motg;
Servo mott;

void setup() {
    motd.attach(6);
    motg.attach(5);
    mott.attach(3);

    motd.writeMicroseconds(1500);
    motg.writeMicroseconds(1500);
    mott.writeMicroseconds(1500);
    delay(3000);
    Serial.begin(9600);

    int xval[10], yval[10], zval[10];
    long xsomme = 0;
    long ysomme = 0;
    long zsomme = 0;

    for(int i = 0; i < 10; i++){
        xval[i] = ar(x);
        yval[i] = ar(y);
        zval[i] = ar(z);
        xsomme += xval[i];
        ysomme += yval[i];
        zsomme += zval[i];
        delay(10);
    }

    xmoy = xsomme / 10.0;
    ymoy = ysomme / 10.0;
    zmoy = zsomme / 10.0;
}

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void loop() {
    xpos = ar(x);
    ypos = ar(y);
    zpos = ar(z);

    if(xpos < xmoy - seuil){
        xout = map(xpos, 0, xmoy - seuil, -400, 0);
    }
    if(xpos > xmoy + seuil){
        xout = map(xpos, xmoy + seuil, 1023, 0, 400);
    }
    if(ypos < ymoy - seuil){
        yout = map(ypos, 0, ymoy - seuil, -400, 0);
    }
    if(ypos > ymoy + seuil){
        yout = map(ypos, ymoy + seuil, 1023, 0, 400);
    }
    if(zpos < zmoy - seuil){
        zout = map(zpos, 0, zmoy - seuil, -200, 0);
    }
    if(zpos > zmoy + seuil){
        zout = map(zpos, zmoy + seuil, 1023, 0, 200);
    }
    if(xpos > xmoy - seuil && xpos < xmoy + seuil){
        xout = 0;
    }
    if(ypos > ymoy - seuil && ypos < ymoy + seuil){
        yout = 0;
    }
    if(zpos > zmoy - seuil && zpos < zmoy + seuil){
        zout = 0;
    }

    pousse(yout, xout, zout);
}

```

```

void pousse(int yout,int xout, int zout){
    int lpow, rpow, tpow;
    if(yout>=0){
        lpow = 1500 - yout - xout;
        rpow = 1500 + yout - xout;
    }
    if(yout<0){
        lpow = 1500 - yout + xout;
        rpow = 1500 + yout + xout;
    }
}

```

```
}
tpow = 1500 + zout;

if(lpow>1900){
    lpow = 1900;
}
if(lpow<1100){
    lpow = 1100;
}
if(rpow>1900){
    rpow = 1900;
}
if(rpow<1100){
    rpow = 1100;
}
motg.writeMicroseconds(lpow);
motd.writeMicroseconds(rpow);
mott.writeMicroseconds(tpow);

Serial.print("MTG: ");
Serial.print(lpow);
Serial.print("  MTD: ");
Serial.print(rpow);
Serial.print("  MTT: ");
Serial.println(tpow);
}
```